SIMBLOCKS // RESULT

2026-10-15 23:20Z

TRIAL 2 WEAPON SHAFT BEND (SYSTEM FILE)

● PASS VERIFIED

Does a 10 mm 4140 prehard dead shaft take a permanent bend over 0.1 mm on a square 600 J hit?

SURVIVES? FAIL

NO

shaft bend SF 0.721

SHAFT BEND [GOVERNING] FAIL

SF 0.721

P(fail) 67% | SF P5 0.053

Margin to each user-defined failure mode



Trial 2 weapon shaft bend (system file): VERIFIED | PASS 13 WARN 1 FAIL 0 SKIP 0 | 169 runs, 144.3 s

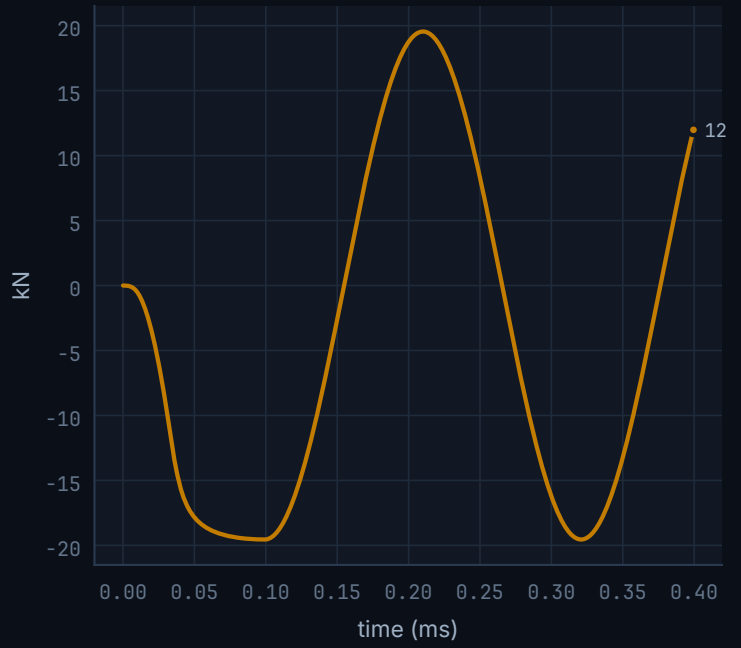
TRIAL 2 WEAPON SHAFT BEND (SYSTEM FILE)

Time histories of the governing states (nominal case)

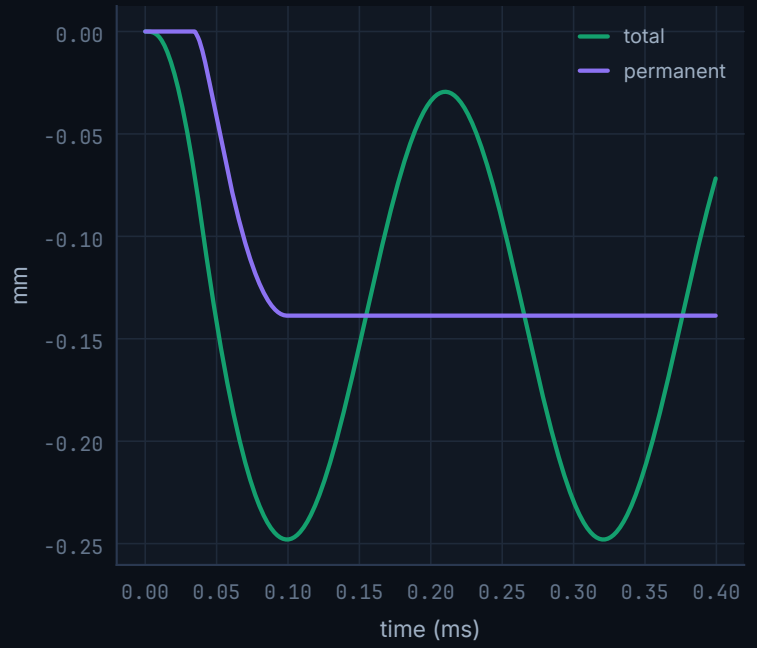
Tooth contact force



Shaft force



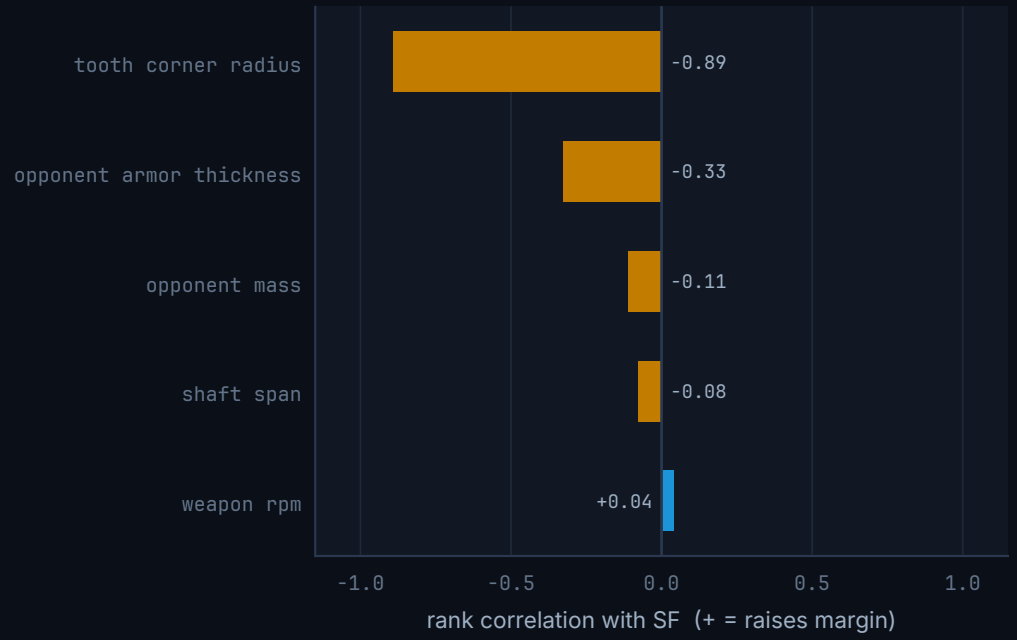
Shaft deflection and permanent set



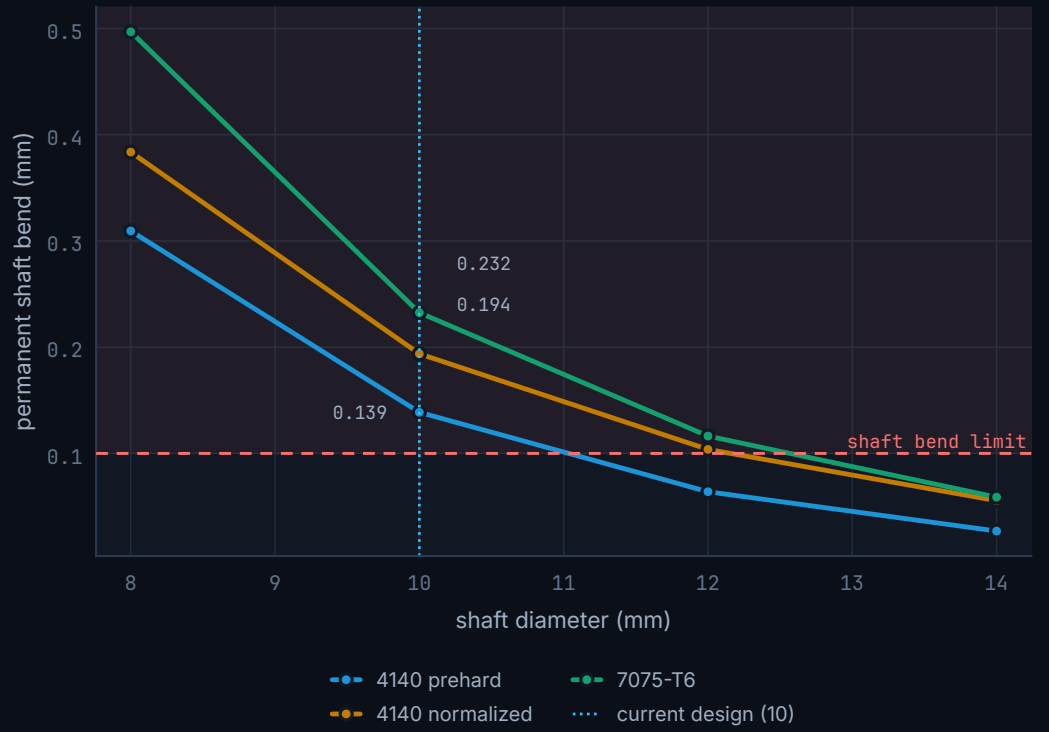
TRIAL 2 WEAPON SHAFT BEND (SYSTEM FILE)

What to design around: drivers of the governing margin and the design-variable sweep

Drivers of 'shaft bend' margin



shaft_set_mm vs d



SIMBLOCKS // VERIFICATION

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TRIAL 2 WEAPON SHAFT BEND (SYSTEM FILE)

● PASS VERIFIED

Automated verification battery

CHECK	STATUS	DETAIL
determinism	● PASS	bit-identical repeat
finite_outputs	● PASS	all outputs finite
physical_bounds	● PASS	3 bounds respected
stability_margin	● PASS	critical dt 2.43e-05 s / dt 2e-08 s = 1214.9x
dt_convergence	● PASS	shaft_set_mm: err 0.066% (order 0.99); dent_mm: err 3.4e-05% (order 4); F_contact_peak_kN: err 0% (order inf); F_shaft_peak_kN: err 0.0042% (order ...
energy_ledger	● PASS	worst closure residual 0.0056% across 3 dt levels (tol 5%)
limit:weapon not spinning → nothing happens	● PASS	matches expected limit
knockout:no tooth_contact	● PASS	shaft_set_mm -100%, dent_mm -100%, F_contact_peak_kN -100%, F_shaft_peak_kN -100%, E_to_opponent_J -100%, rpm_after +18.5%, perforated -100%
knockout:tooth_contact elastic only	● PASS	shaft_set_mm +9.24e+03%, dent_mm -100%, F_contact_peak_kN +1.22e+03%, F_shaft_peak_kN +22.4%, E_to_opponent_J +8.28e+03%, rpm_after -151%, perforat...
knockout:no shaft	● PASS	shaft_set_mm -100%, dent_mm +0.00413%, F_contact_peak_kN +0%, F_shaft_peak_kN -100%, E_to_opponent_J +0.455%, rpm_after -0.0421%, perforated +0%
knockout:shaft elastic only	● PASS	shaft_set_mm -100%, dent_mm +0.00057%, F_contact_peak_kN +0%, F_shaft_peak_kN +83.1%, E_to_opponent_J +0%, rpm_after +0%, perforated +0%
sensitivity	● PASS	5 inputs swept low/high (trends reported, not judged)
monte_carlo	● PASS	150 samples, 0 failed finite/bounds/ledger (0.0%)
failure_modes	● WARN	shaft bend: nominal SF 0.721 FAILED, P(fail) 66.7%